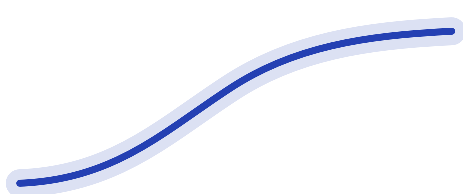
Modelling how children with Down syndrome learn language

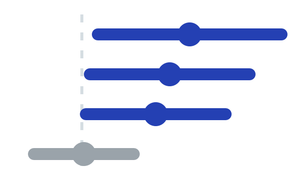
Two studies: how vocabulary grows, and what reading-and-language teaching causes.



Study 01

Vocabulary growth

How children understand, say and sign words as they grow, across fourteen datasets.



Study 02

Reading & language

A re-analysis of a randomised trial, separating what the teaching causes from what tracks ability.

Ethan Buckley · MSci Natural Sciences, UCL

Bayesian hierarchical modelling · causal inference · Python

One method, two studies

Both studies run on the same Bayesian foundation. What changes is the question, and how far the data can be pushed toward cause.

Study 01 · Vocabulary growth

Vocabulary growth

How do children build words?

Models the words children **understand, say and sign together**, pooled across fourteen datasets. The comprehension–production gap falls straight out of the model as the production ratio **q**.

What's distinctive

The understood × production-ratio decomposition, estimated jointly across fourteen datasets.

Study 02 · Reading & language

Reading & language

What does teaching cause?

Re-analyses a **randomised trial** through a locked causal DAG, so the teaching effect is identified by randomisation. Gradient-boosting discovery feeds the Bayesian estimates.

What's distinctive

A causal DAG plus randomisation; a two-layer discovery-then-estimation method.

The shared foundation

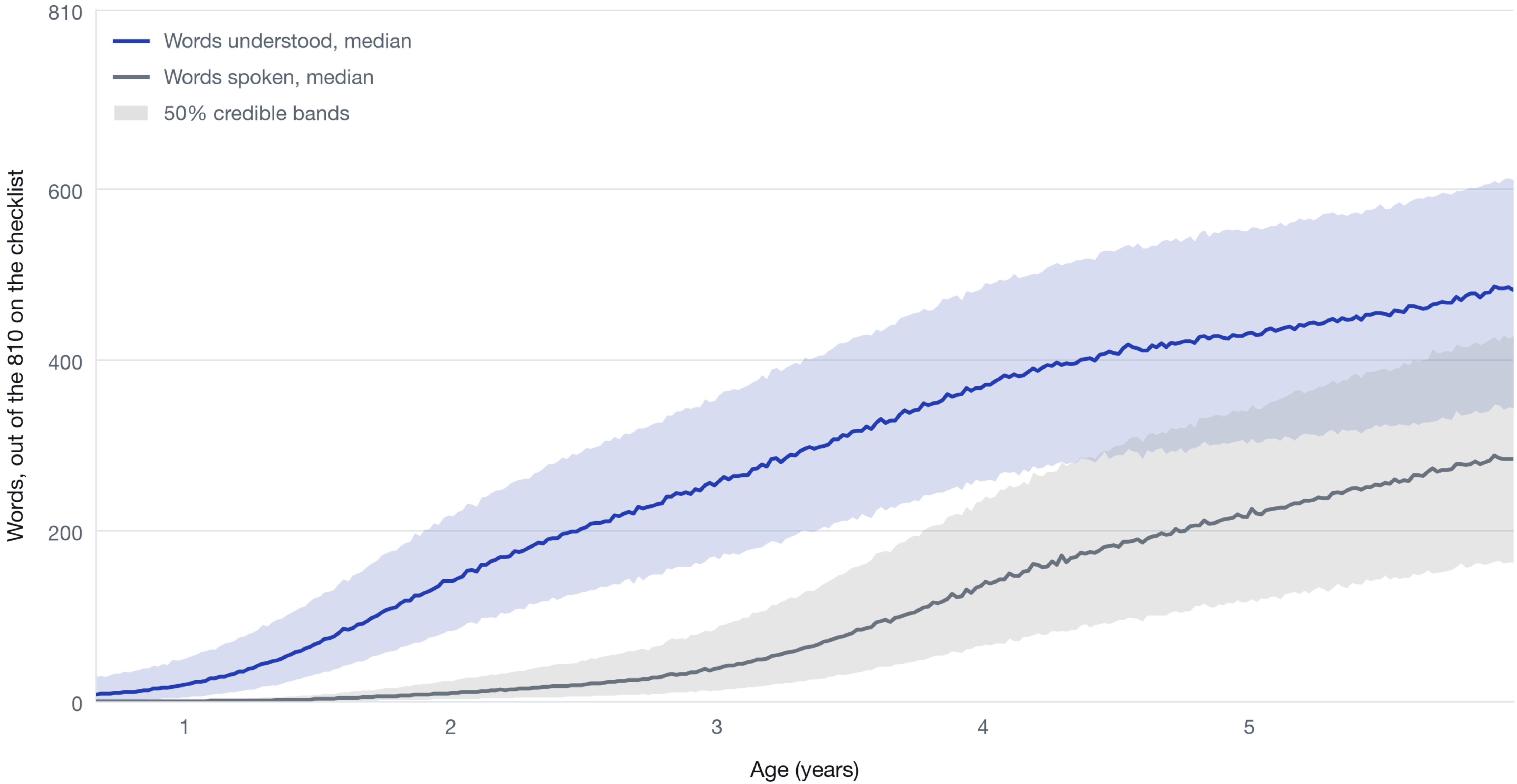
`Hierarchical Bayesian models` `Beta–Binomial likelihood` `Gaussian-process (HSGP) trends` `credible intervals throughout`

`reproducible: PyMC · NUTS · DuckDB · Quarto`

The same engine underneath: **full-posterior inference**, checked with predictive checks and cross-validation, and reproducible from raw data to report.

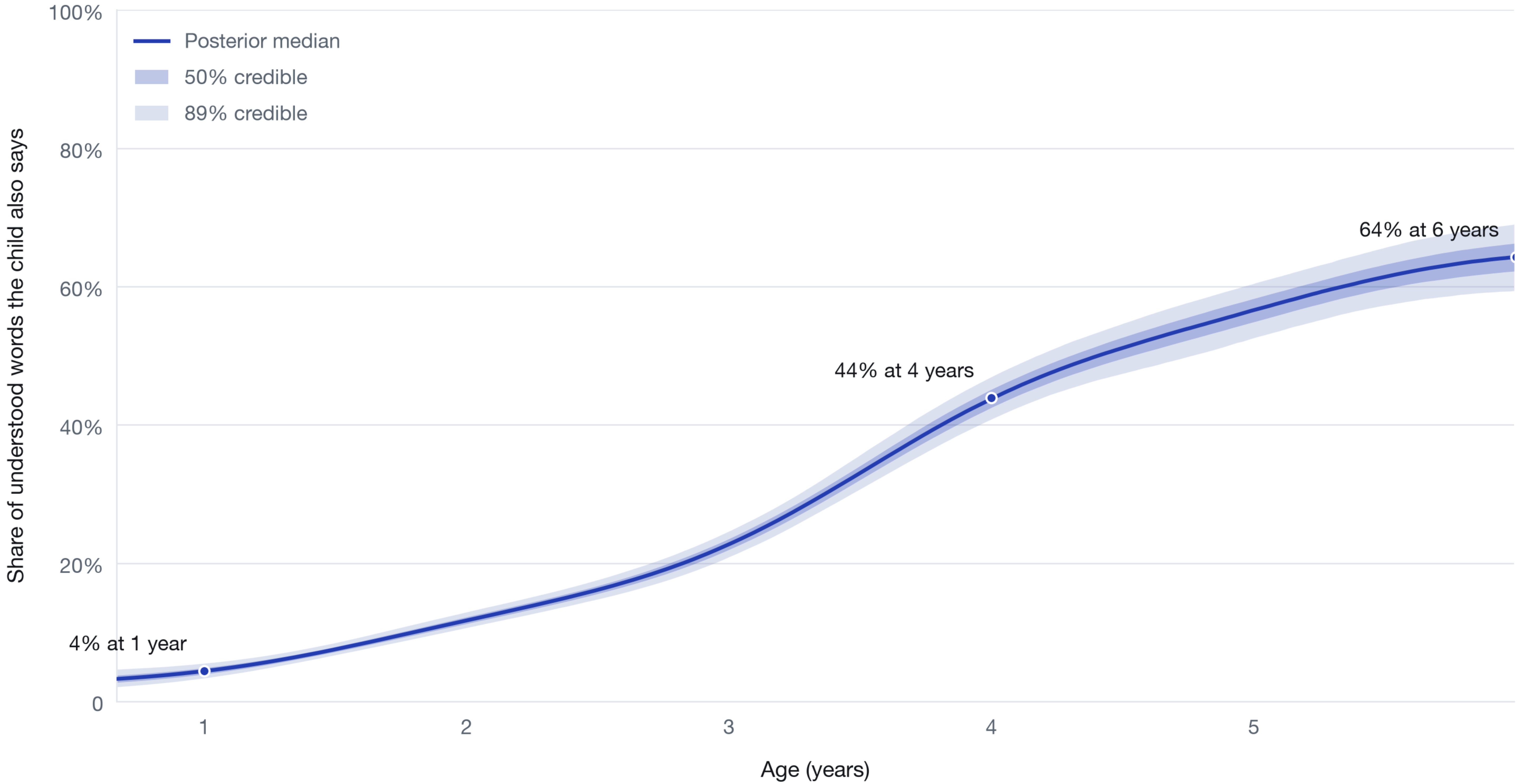
Words understood and words spoken by age, children with Down syndrome

Posterior-predictive medians with 50% credible bands. The model estimates spoken vocabulary as words understood multiplied by the share also said, so speaking can never exceed understanding, and the gap between the curves is the comprehension-production gap.



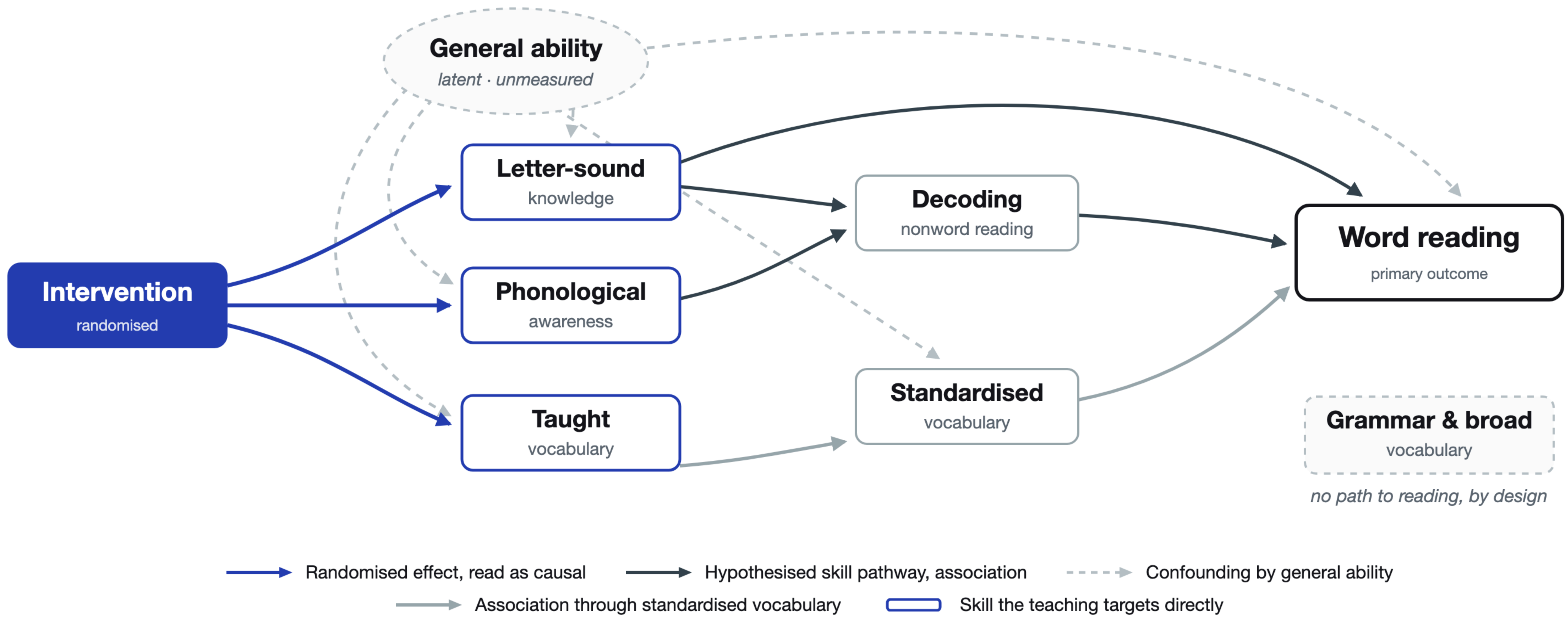
Share of understood words that children with Down syndrome also say, by age

Posterior median with 50% and 89% credible bands, from a hierarchical Bayesian model fitted to 1,424 parent-report vocabulary assessments pooled from 14 datasets in six countries.



Causal structure assumed in the reading analysis

Randomisation makes the intervention independent of general ability, so its effects on the skills it targets are read as causal. Every skill-to-skill link stays confounded by ability and is reported as association, not cause.



Effect of the reading intervention on each skill

Intention-to-treat estimates on the log-odds scale with 89% equal-tailed intervals; 54 children randomised to the teaching or a waiting list. Right of zero favours the teaching. Blue intervals exclude zero; grey intervals span it.